

Product Name: AI Fitness Coach (with Admin Panel)

1. Overview

AI Fitness Coach is an AI-powered platform delivering personalized diet and workout plans using **body analysis, user goals, and continuous tracking**, now extended with a **robust Admin Panel** for monitoring, control, and optimization.

2. Objective

- Deliver personalized fitness guidance
 - Improve user consistency with AI + tracking
 - Enable admins to **monitor users, manage AI outputs, and maintain quality**
-

3. User Roles

3.1 End User

- Uses fitness features (plans, tracking, chatbot)

3.2 Admin

- Monitors platform activity
 - Manages users & plans
 - Controls AI outputs & moderation
-

4. Core Features (User Side)

4.1 AI Onboarding (Body Analysis)

- 4 image upload (Front, Back, Left, Right)
- MediaPipe:
 - Posture detection

- Body landmarks
 - Estimated BMI
-

4.2 Goal Selection

- Weight Loss / Gain / Muscle / Maintenance
-

4.3 AI Diet Plan Generator

- Personalized meals
 - Calories + macros
 - Allergy-aware
-

4.4 Workout Plan Generator

- Home/Gym plans
 - Sets, reps, weekly split
-

4.5 Daily Habit Tracker

- Meals, water, workout, sleep
 - Streak system
-

4.6 AI Chatbot (RAG)

- Context-aware answers
 - Uses:
 - User plan
 - Progress data
-

4.7 Weekly Progress Tracking

- Photo comparison
 - AI insights
-

4.8 Dashboard

- Weight, calories, streaks
 - Fitness score
-

5. Admin Panel (NEW - ADVANCED)

5.1 User Management

- View all users
 - Search/filter users
 - Ban / deactivate accounts
 - View user activity:
 - Last login
 - Plan usage
 - Progress stats
-

5.2 Analytics Dashboard

- Total users
 - Active users (DAU/WAU)
 - Plan completion rates
 - Chatbot usage metrics
 - Avg. user fitness score
-

5.3 AI Output Monitoring

- View generated diet/workout plans
 - Flag inappropriate or inaccurate outputs
 - Adjust prompt templates (admin-controlled AI tuning)
-

5.4 Image Moderation

- Review uploaded body images
 - Detect misuse/inappropriate uploads
 - Delete flagged content
-

5.5 Plan Management (Override System)

- Admin can:
 - Edit AI-generated plans
 - Create custom templates
 - Assign manual plans to users
-

5.6 Chat Moderation

- View chatbot conversations
 - Detect harmful queries
 - Block abusive users
-

5.7 Reports & Logs

- System logs
 - AI usage logs (API calls, tokens)
 - Error tracking
-

6. AI Components

6.1 Computer Vision

- MediaPipe:
 - Pose detection
 - Body structure estimation

6.2 LLM (OpenAI / Gemini)

- Diet/workout generation
- Chatbot

6.3 RAG System

- Stores:
 - User plans
 - Progress history
 - Enables personalized responses
-

7. Technical Architecture

Frontend (React)

- User Dashboard
- Admin Dashboard (separate routes)
- Chat UI / Material UI / ShadeCN / Tailwind CSS
- Image upload

Backend (Node + Express)

- REST APIs
- Auth (JWT + Role-based access)

Real-Time

- Socket.IO:
 - Chat updates
 - Live tracking sync

Database (MongoDB)

Collections:

- Users (role: user/admin)
- Plans
- Progress
- Chat logs
- Admin logs

AI Layer

- OpenAI / Gemini
 - Prompt management system
-

8. Role-Based Access Control (RBAC)

- User → limited access
 - Admin → full control
 - Middleware:
 - verifyJWT
 - checkRole('admin')
-

9. User Flow

1. Signup/Login
 2. Upload body images
 3. AI analysis
 4. Select goal
 5. Get plan
 6. Track daily habits
 7. Chat with AI
 8. Weekly updates
-

10. Admin Flow

1. Admin login
 2. View dashboard analytics
 3. Monitor users
 4. Review AI outputs
 5. Moderate content
 6. Adjust system settings
-

11. Constraints

- AI not medically certified
 - Image analysis is approximate
 - Requires moderation for safety
-